DATA ANALYTICS

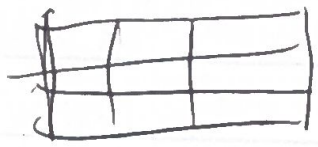
Data is Information collected that is true.

First Party data
(Data you collected by yourself) or you generate yourself

3rd Party
Data you get from someone else

forms of Data

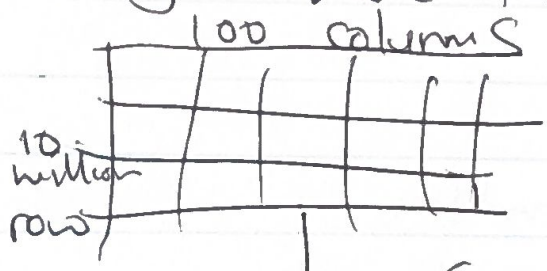
Structured Data
(Organised data)
(Data with a known formate or structed format)
~~Data with a structed~~
Data in tables such as excel



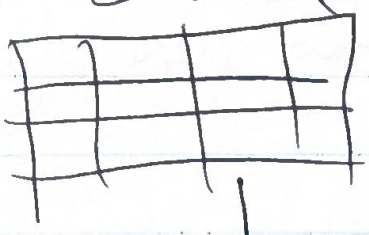
Unstructed data
- Data without a forma
- No specific format

eg music, video, pho call, voice note, PDF document photos

Kiky Data (structed data still have a problem
The Data might have too much information

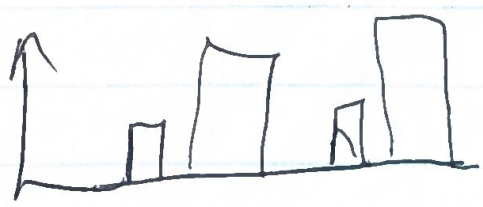


(we use it to clean structed data) SQL (structed Query Language)



(Aggregate it. Summarise what you need

Insights



Powerful Excel

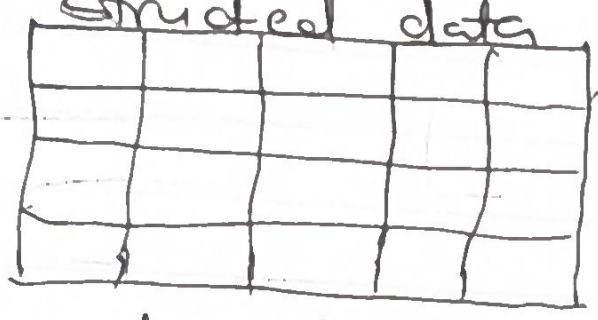
What do we use SQL for
What is aggregated Data
What is Power BI

① SQL ~~Wasnd~~ Stand for Structed Query Language
* Is a programming language for storing and processing information in a relational database. (store, retrieve, update and manage data.
* A database is an organized place where information is stored.
* A database is an organized place where information is stored so it can be easily found, used and updated
* A database is like a digital filing cabinet instead of papers, It stores data on a computer.

② Aggregated data means data that has been collected ~~instead of~~ and combined into a summary.
* Instead of looking at individual details aggregated data shows the total, average, count or summary.

LESSON 1

Introduction to SQL



Big Data

→ Structed Query language (SQL)

→ Data Clearing
Data processing
Data aggregation

Lessons 1
Introduction to SQL



Summarized Data

Excel
PowerBI
- Insights
- story telling



Database
Data warehouse
Big Data

→ DMS SQL →

Data bricks
snowflake
Microsoft SQL Server

summarised Data

- Raw Data
- Data source for Analysis

Data Clearing
→ Data Aggregation

- PowerBI
- Excel
- looker studio

Data Visuals

SQL Syntax — the way we write SQL

SELECT statement

- It allows to specify the columns that we want from a specific table.

FROM statement

- you specify the name of the database . Schema . table name

~~Database Schema~~

Select

Column 1

Column 2

Column 3

from Database . Schema . table . name .

Cyrro . students . maths - marks

Database — name → Laptop

Schema — name → School

table — name → Results .

Curro . Students . maths - marks

* SELECT DISTINCT (Different values in that column)

FROM database . schema . table - name

Lesson 2

SQL Filtering Statement

Where statement

* It allows us to specify condition in order to filter rows in our table
* We are filtering the rows

NB, We filter the columns using the select statement
We filter the rows using the where statement

Syntax: SELECT Column 1, { on the last column before the from statement you don't put comma }
Column 2,
Column 3,
Column 4

FROM database . Schema . table
Where condition

database schema table
Curro . Students . marks

Student ID	Name	Subject	Marks
101	Rafhi	Maths	99
102	Mbali	Maths	22
103	Fifi	Geo	50
104	Tumelo	Bio	40

Question 1
Show me the student with marks less than 50

Select , Name , marks
From Curro . Students . marks
Where marks < 50;

Question 2
Write an SQL script to give me the student ID name and marks of students who got above 50

SELECT : student ID , name , marks
FROM Curro . students . marks
Where mark > 50 ;

student ID	Name	Marks
101	Rafhi	99

SQL Comparison operators

Operator	Description
=	Equal to
>	Greater than
>=	Greater than or equal to
<	less than
<=	less than or equal to
<>	not equal to
!=	

We use the comparison operators in the WHERE clause in order to filter data

SELECT * → star means select all the columns in the table
FROM Curro.students.marks
WHERE subject = 'Geo';
We use single quotes because Geo is a word not a number.

LESSON 3
SQL Operators
* We use the operators as part of our filtering statement in order to build the condition for filtering

SQL Comparison operators
SQL Logical Operators
AND operator
* Is a statement that help ~~of~~ to join multiple conditions
* Both conditions must be met or satisfied

SELECT Column 1,
Column 2,
Column 3
FROM database . schema . table
WHERE Condition 1 AND Condition 2

Curro . students . marks

student ID	Name	Subject	Marks
101	Rofhi	Maths	99
102	Mbali	Maths	22
103	Lerato	Geo	50
104	Ally	Acc	40

Question
We want the name and marks of a student who got above 50 %

SELECT name,
marks
FROM Curro . students . marks
WHERE ~~marks~~ marks > 50 AND Subject = 'Math' AND Marks > 50

Name	Marks
Rofhi	99

OR (operator) statement
is a statement that allows us to specify multiple condition but at least one condition must be met

SELECT column 1,
column 2
FROM database.schema.table
WHERE condition 1 OR condition 2

Question: Any one doing maths or they have marks greater or equal to 50.

SELECT *
FROM Curro_students.marks
WHERE Subject = 'maths' or marks >= 50

Student ID	Name	Subject	Marks
101	Rofhi	Maths	99
102	Mbali	Maths	22
103	Lerato	Geo	50

LESSON 4
Logical operator
Between operator
It allows us to filter our data on a range of values.
It is an inclusive operator

SELECT column 1,
column 2
FROM database.schema.table
WHERE column name Between min AND max

Question
Write an SQL query to get the name, subject and marks of students who obtain marks between 0 and 30

```
SELECT  Name,
        Subject
        Marks
FROM    Curro.students.marks
WHERE   Marks BETWEEN 0 AND 30;
```

Name	Subject	Marks
Mkalti	Maths	22



Power BI
Excel

Power Point

Question

Write an SQL query to set the name, subject and marks of students who obtain marks between 50 AND 100

```
SELECT  name,
        Subject,
        Marks
FROM    Curro student marks
WHERE   marks BETWEEN 50 AND 100;
```

Student ID	Name	student	Marks
101	Rachi	maths	99
103	lerato	Geo	50

IN Statement

Allows us to specify specific values in a column that we want to include

The values not included in will be excluded from the final outcome

```
SELECT    column1,
          Column2
FROM      database.schema.table
WHERE     column  IN (value 1, Value 2, Value 3);
```

Shoprite.online.sales

Customer id	Customer name	City
111	Rofhi	JHB
112	Mbali	CPT
113	Lerato	JHB
114	Thabo	PTA

```
SELECT    customer_name,
          City
FROM      Shoprite.online.Sales
WHERE     city IN ('JHB', 'PTA')
```

Custome name	City
Rofhi	JHB
Lerato	JHB
Thabo	PTA

Lesson 6

Sorting Statements

It helps us to arrange our data from either the smallest to the biggest or from the biggest to the smallest.

we order our data using a specific column.

ORDER BY:

- Ascending order (ASC)
  - smallest to biggest
  - new to old
  - a - z
  - 1 - 9

```
SELECT columns
FROM table
WHERE condition
ORDER BY column (ASC)
```

- Descending order (DESC)
  - Biggest to smallest
  - old to new
  - z - a
  - 9 - 1

```
SELECT columns
FROM table
Where condition
ORDER BY column (DESC)
```

Products

Product id	Product name	Price
111	Apple	2
112	Banana	4
113	Orange	3
114	Kiwi	5

```
SELECT Product name,
       Price
FROM products
ORDER BY price DESC;
```

To rearrange your data in a specific order

Product Name	Price
Kiwi	5
Banana	4
Orange	3
Apple	2

SQL
Ascending order:

```
SELECT product name,
       price
FROM products
ORDER BY Price Ascending
```

Product Name	Price
~~Kiwi~~ Apple	2
Orange	3
Banana	4
Kiwi	5

LIMIT (FUNCTION) Statement:
It limit the number of raws that you want to display when you run code. (when running a specific query)

Question:
When you want to see only 2 raws

```
SELECT *
FROM products
LIMIT 2;
```

Product ID	Product Name	Price
111	Apple	2
112	Banana	4

Nicknames

SQL Aliases (means to give a new name to a column)

We use the aliases to rename the column name or the table name within the SQL Script

We use aliases (AS) to make our column names to be more descriptive.

```
SELECT product_id (AS) barcode,
       product_name AS fruit_name,
       Price
FROM products;
```

Barcode	fruit_name	Price
111	Apple	2
112	Banana	4
113	Orange	3
114	Kiwi	5

Aggregate functions

Consolidate our data, Summarise our data, combine into groups.

Aggregate functions summarise many rows into one result (or one result per group)

1. COUNT() → how many rows (Summarise)
2. SUM() → generate the total of a numeric column
3. Avg() → Average of a numeric column
4. Min() → Smallest value
5. Max() → Largest value

- An aggregate looks at multiple rows in a table and return a single value
- An aggregate function is used to perform a calculation in the rows of our table in a specific column and returns a single value.

Aggregate functions

* What is an aggregate function?
* It summarizes data into one result
- Is a fuction that performs a calculation a multiple rows of data and returns a single result/value.
- Perform calculation in our rows and returns one result or one result per group

Aggregate functions are commonly used with the GROUP BY statement to group rows that belong to the same category

```
SELECT Aggregate function (column1) AS descriptive name
       Categorical - column
FROM   table_name
GROUP BY Categorical_column;
```

- We only use the GROUP BY statement when we are aggregating the data and in our SELECT we have a Categorical column
- We do not GROUP BY the aggregate column

What is a categorical column?
Is a column that give us a classification of specific data eg gender
It differentiate the data into different groups

COUNT( ) - we commonly count the unique identifiers (ID)

```
SELECT COUNT (ID) AS Number of students,
FROM grade 10 11
GROUP BY gender ;
```

- You cannot GROUP BY without having a categ
column in the SELECT

```
SELECT MAX (Revenue) AS Highest_revenue
FROM products;
```

Highest_revenue
1.5

```
SELECT MIN (Revenue) AS Lowest_Revenue
FROM products;
```

Lowest_Revenue
2

```
SELECT COUNT (ID) AS number of of fruits
FROM Products;
```

number of fruits
4

GROUP BY

GRADE	GENDER
M	M
F	M
M	F
F	M

ID	Name	Gender
101	Mbali	F
102	Raffi	M
103	Thabo	M
104	Jacob	M
105	John	M
106	Lerato	f
107	Robert	M
108	Anna	F

Exercise

Sale_Id	Store	Amount
1	Mr Price	100
2	Mr Price	50
3	Prep	80
4	Pep	20

Question

① Write a SQL query to calculate the total revenue generated by all stores

② Write a SQL query to calculated the total revenue per store group

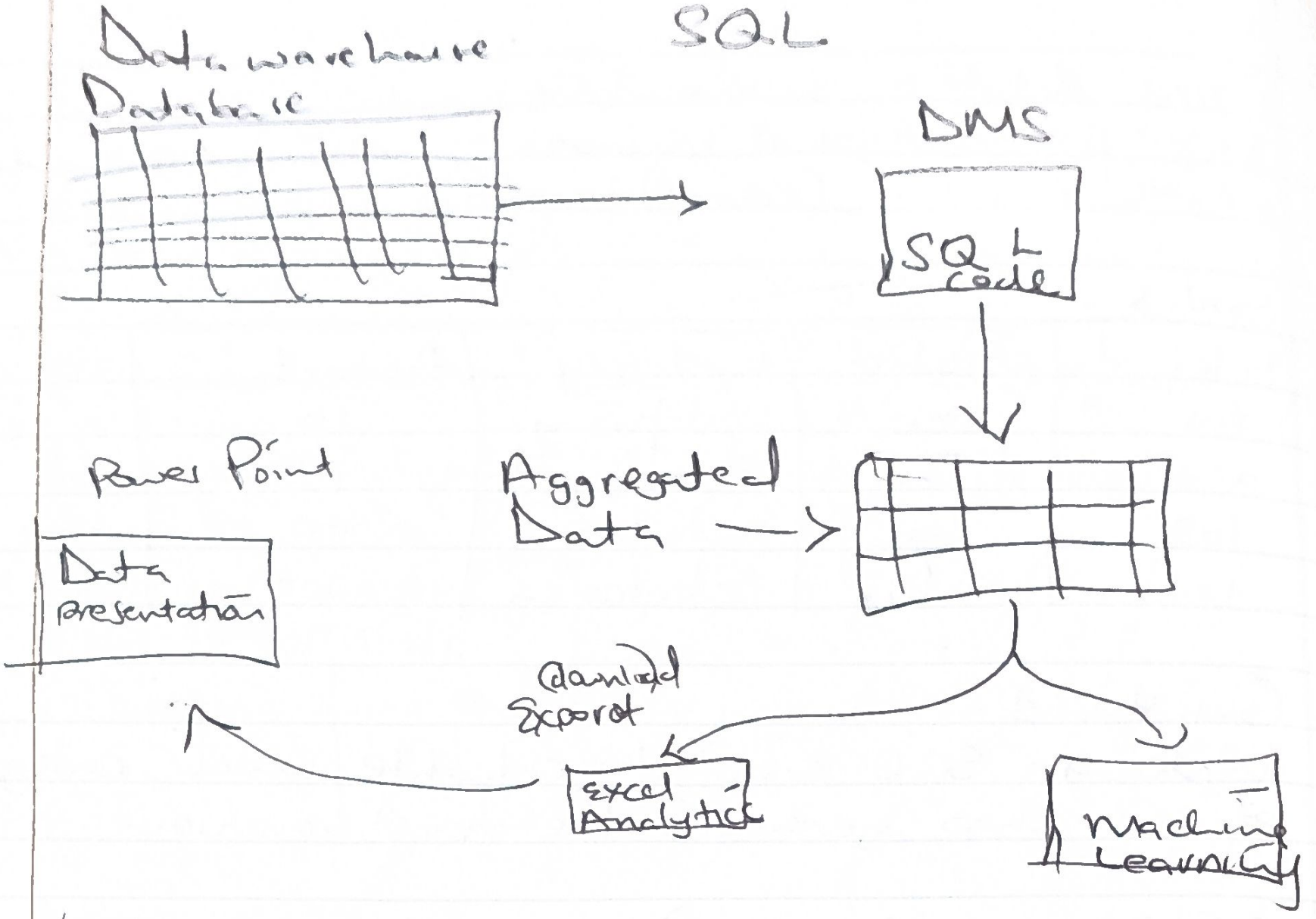
① SELECT SUM(amount) AS total revue revenue
FROM Sales;

total Revenue
250

② SELECT store, SUM(amount) AS total revenue
FROM sales
GROUP BY store;

NB Do not perform SUM() on an ID or a unique identifier

Store	total_revenue
Mr Price	150
Pep	100



LESSON 10

Aggregate Functions

① SUM ( )
② COUNT ( )
③ MIN ( )
④ MAX ( )
⑤ AVG ( )

performs a calculation and returns a single value

rows in a column

we put the column name

GROUP BY
we group data into categories

HAVING
help us to filter data that is grouped

Sitax of HAVING

SELECT AGG. FUN (column) AS new column. name,
Categorical. column

Sales

Sale id	Product	Category	Amount
1	Pen	Stationery	10
2	Pen	Stationery	15
3	Book	Stationery	40
4	Mouse	Electronics	200

Question 2

1) Write an SQL query to calculate total revenue for each category.

```
SELECT SUM(Amount) AS total revenue
       Category
FROM   Sales
GROUP BY Category
```

```
SELECT category,
       SUM(amount) AS total revenue
FROM   Sales
GROUP BY category
```

Category	total revenue
Stationery	65
Electronics	200

```
SELECT  Category,
        SUM(amount) AS Revenue
FROM Sales
GROUP BY Category;
```

Category	Revenue
Stationery	65
Electronics	200

A category that generates more than R100

HAVING - we filter data that is aggregated

ID	Name	Department	Salary
1	Ropfiwa	IT	100
2	Mbali	HR	20
3	Lerato	HR	10
4	Sipho	Sales	15
5	[illegible]	Sales	40

```
SELECT  department,
        SUM(Salary) AS total Salary
FROM employees
GROUP BY department;
```

Department	total - Salary
IT	100
HR	30
Sales	25

Logical Operators to build specific logic
AND - Both conditions must be met
OR - One condition must be met
IN - Specify values you want
BETWEEN - Inclusive statement
NOT - Gives opposite result
- Can be used in the WHERE statement when filtering data

Aggregate Functions
- Summarise data
- Consolidate data
- Combined data into groups
- Performs calculation on multiple rows and returns a single value per group

FUNCTIONS
① MIN () ③ COUNT () ⑤ SUM ()
② MAX () ④ AVG ()
- These aggregate functions are used with the GROUP BY statement
- HAVING statement is used to filter aggregated data

ORDER BY is sorting statement used to arrange the data either in Ascending order or Descending order

ASC - Ascending Order is Small to big 1 - 9, A - Z
DESC - Descending order is big to small 9 - 1, Z - A

CASE statement
- Helps us to create flags in our data
- Build logical conditions

- Creates a new column
- Placed within the SELECT statement
- Creates logic
- Creates a categorical column from a numeric column

CASE
    WHEN condition1 THEN 'result1'
    WHEN condition2 THEN 'result2'
ELSE 'default result'
END AS new_column_name

→ Given when condition1 and condition2 are not met

Aggregated Fuctions

- Count how many rows in a table

SELECT COUNT(*) AS number of rows
FROM table;

Counts the number of rows in a/the table

- We also use the COUNT() to count the unique Identifier
- COUNT (DISTINCT ID)
  used to count the number of different values in a specific column

ID
101
102
101
103

SELECT COUNT (DISTINCT ID)
FROM employees;

3

SUM ( ) - You can not sum the ID
- You can not SUM words
you can sum Salary
Amount
Price
Revenue

SUM () means addition

GROUP BY you GROUP BY the column that is not aggregated

```
SELECT
        Categorical_column1,
        AGG_FUN (column2) AS descriptive_column_name
FROM    table_name
GROUP BY  Categorical_column;
```

SQL JOINS

① Primary key → is a column that uniquely identify each row
- no duplicates

Customers

Id customer	Name
1	Raffaela
2	Leardo

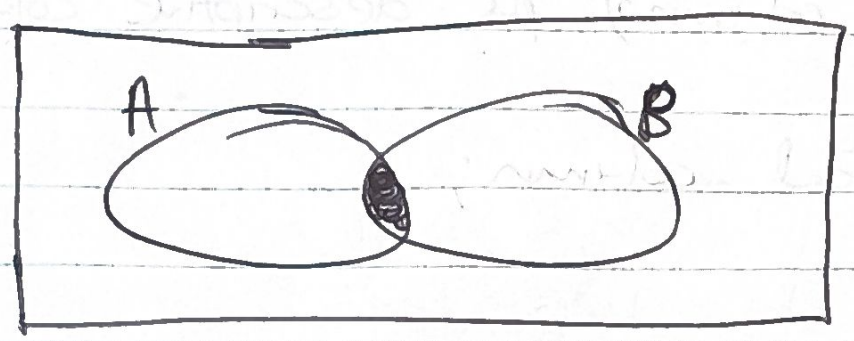
② Foreign key →

Orders

Orders Id	Customer Id	Amount
101	1	
102	1	
103		

INNER JOIN

Allows us to join two or more tables on a common column then give us the rows that are matching in both tables



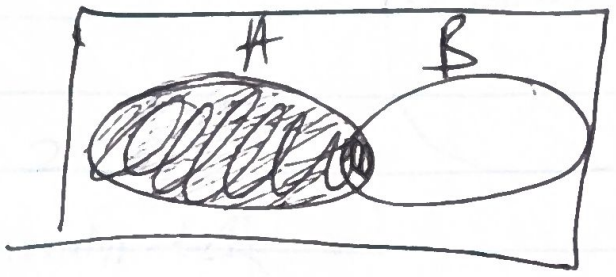

```
SELECT  column 1,
        Column 2,
        Column 3
FROM    table 1  AS A
INNER JOIN table 2 AS B
ON A.common_ column = B.common_column;

SELECT  customer id,
        name,
        Amount
FROM    Customers  As A
INNER JOIN  orders  AS B
ON A. Customer id  = B. customer id ;
```

Customer id	Name	Amount
1	Rofhiwa	300
1	Rofhiwa	150

LEFT JOIN

- Returns everything from the left table (base table) and adds matches from the right table.
- If no match right side becomes NULL




```
SELECT Column 1,
       Column 2
       Column 3
FROM table 1 AS A          ← Left table
LEFT JOIN table 2 AS B     ← Right table
ON A. common column = B. common . column .
```

```
SELECT Customer id,
       name,
       Amount
FROM customers AS A
LEFT JOIN orders AS B
ON A. Customer id = B. Customer id;
```

Customers id	Name	Amount
1	Rofhiwa	300
1	Rofhiwa	150
2	Lerato	NULL